

Hypothesis Testing II

Lecture 13

Topic in this lecture

- Chi-square test

Chi-Square Test

- In some situation, we want to test whether the observed values from an experiment and the expected values are likely to be close to each other
- Thus, we will construct a test statistics that measures the closeness of the observed to the expected values. The statistic is called the chi-square statistic
- Let k be the number of outcomes, and let O_i and E_i be the observed and expected number of trials, respectively, that result in outcome i . The chi-square statistic is

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Chi-Square Test (2)

- The larger the value of χ^2 , the stronger the evidence against H_0 .
- To determine the P-value for the test, we must know the null distribution of this test statistic. In general, we cannot determine the null distribution exactly. However, when the expected values are all sufficiently large, a good approximation is available.
- The distribution is called chi-square distribution with $k-1$ degrees of freedom
- The use of chi-square distribution is appropriate whenever all the expected values are greater than or equal to 5

Case Study 1

- A gambler wants to test a 4-sided die to see whether it deviates from fairness. Let p_i be the probability that the number i comes up. The null hypothesis will state that the die is fair, so the null hypothesis is

$$H_0 : p_1 = p_2 = p_3 = p_4 = 1/4$$

- The gambler rolls the die 1000 times and obtains the results as shown

Category	Observed	Expected
1	260	250
2	238	250
3	245	250
4	257	250
Total	1000	1000

Compute test statistics

Category	Observed	Expected
1	260	250
2	238	250
3	245	250
4	257	250
Total	1000	1000

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Compute P-value

- Look up P-value from χ^2 table at d.f. = $k-1$

Example 1

- The length specification of mouse pads is 12.0-12.3 cm. It is thought that 90% of mouse pads manufactured meet the specification, while 5% are too short and 5% are too long. In a random sample of 1000 mouse pads, 860 met the specification, 60 were too short and 80 were too long. Can you conclude that the true percentages differ from 90%, 5%, and 5%?
 - State the appropriate null hypothesis
 - Compute the expected values under the null hypothesis
 - Compute the value of chi-square statistic
 - Find the P-value. What do you conclude?

Chi-square Test for Homogeneity

- Previously, we test the null hypothesis that the probabilities of the outcomes from a multinomial trial are equal to a pre-specified set of values. Sometimes several multinomial trials are conducted, each with the same set of possible outcomes.
- The null hypothesis is that the probabilities of the outcomes are the same for each experiment

Case Study 2

- Three machines are used to manufacture tetrahedral dice. A gambler wants to test if the dice from these 3 machines are the same.

	1	2	3	4	Total
Machine A	98	89	116	97	400
Machine B	100	103	110	87	400
Machine C	144	155	135	166	600
Total	342	347	361	350	1400

Hypothesis Setup

- Set null hypothesis

H_0 : For each column j , $p_{1j} = p_{2j} = p_{3j}$

H_1 : For each column j , p_{ij} are not equal

	1	2	3	4	Total
Machine A	98	89	116	97	400
Machine B	100	103	110	87	400
Machine C	144	155	135	166	600
Total	342	347	361	350	1400

Compute Expected Values

- Expected values

$$E_{ij} = \frac{O_{i.} \cdot O_{.j}}{O_{..}}$$

$$E_{11} = \frac{400 \times 342}{1400} = 97.714$$

	1	2	3	4	Total
Machine A	98	89	116	97	400
Machine B	100	103	110	87	400
Machine C	144	155	135	166	600
Total	342	347	361	350	1400

$O_{1.}$ $O_{2.}$ $O_{3.}$ $O_{4.}$
 $O_{.1}$ $O_{.2}$ $O_{.3}$ $O_{.4}$

Expected Values Table

- Expected values $E_{ij} = \frac{O_{i.} O_{.j}}{O_{..}}$

	1	2	3	4	Total
Machine A	97.714	99.143	103.143	100	400
Machine B	97.714	99.143	103.143	100	400
Machine C	146.571	148.714	154.714	150	600
Total	342	347	361	350	1400

Compute Test Statistics

$$\chi^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Compute P-value

- Compute P-value at d.f. = $(I-1)(J-1)$

Compute P-value

- Compute P-value at d.f. = $(I-1)(J-1)$